Lab No. 02

CLO1

Aim: Concept of LAN and WAN

Different network topologies using hubs, switches and router (with

default configurations) and simulations (unicasting and broadcasting)

Topologies covered: Bus, Star, Ring and Mesh

Networks

A **network** is a system of two or more connected devices, such as computers, printers, or servers, that share resources, files, applications, and communication. Networks are categorized primarily by their geographical area and scope:

- **LAN (Local Area Network):** Covers a small area such as a home, office, or school.
- **MAN (Metropolitan Area Network):** Spans a city or a large campus.
- **WAN (Wide Area Network):** Covers a very large geographical area, often connecting multiple LANs across cities or countries (e.g., the Internet).

Networks can be **wired** (using Ethernet cables) or **wireless** (using Wi-Fi). They are essential for enabling collaboration, centralized storage, internet access, and efficient communication between users and devices.

Local Area Network(LAN)

LAN (Local Area Network) is a type of computer network that connects devices (such as PCs, laptops, printers, and servers) within a **small geographical area**, like:

- **Home**
- **Office**
- **School**
- **University campus**.

It allows connected devices to **share resources** (files, printers, internet connection) and **communicate** with each other at **high speed**.

Building the LAN – Connecting the Hosts to Switches
Add three Hosts.

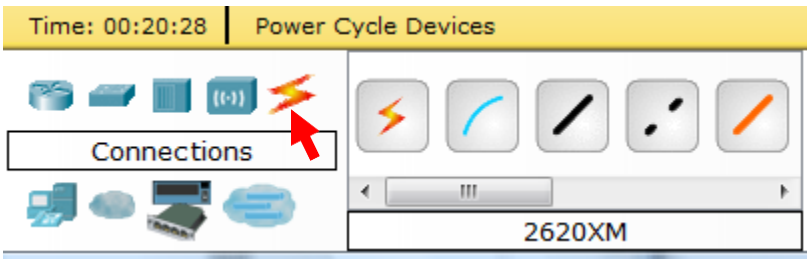


Adding a Switch

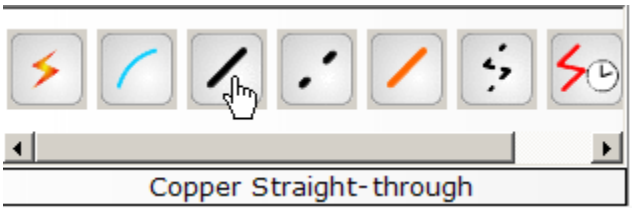
Select a switch, by clicking once on **Switches** and once on a **2950-24** switch.



Connect PCS to Switch0 by first choosing **Connections.**



Click once on the **Copper Straight-through** cable.



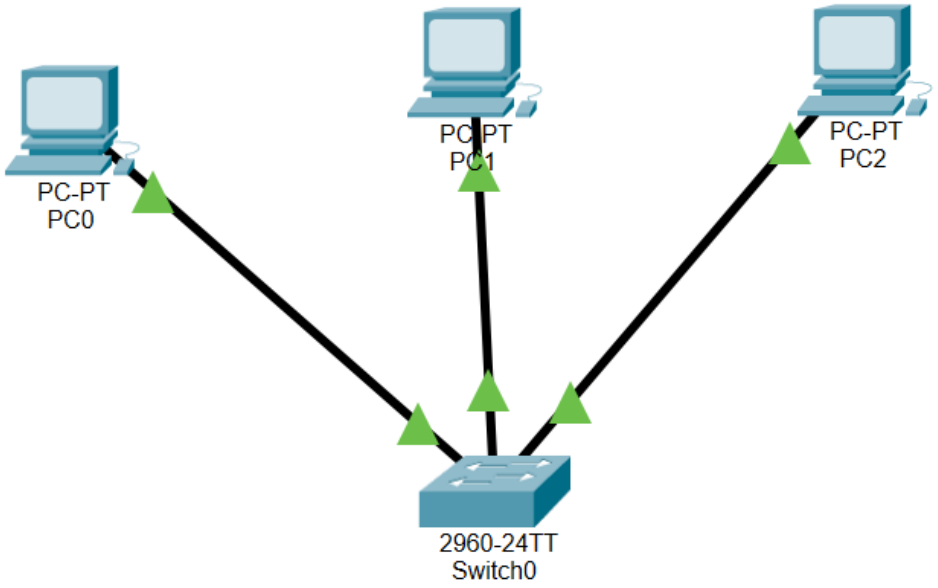
Perform the following steps to connect **PC0** to **Switch0**:

1. Click once on **PC0**
2. Choose **Fast Ethernet**
3. Drag the cursor to **Switch0**
4. Click once on **Switch0** and choose **FastEthernet0/1**
5. Notice the green link lights on **PC0** Ethernet NIC and amber light **Switch0**

FastEthernet0/1 port. The switch port is temporarily not forwarding frames, while it goes through the stages for the Spanning Tree Protocol (STP) process.

6. After a about 30 seconds the amber light will change to green indicating that the port has entered the forwarding stage. Frames can now forward out the switch port.

Repeat the steps above for **PC2 and PC3** connecting it to Switch**0** on port **FastEtherent0/2**. (The actual switch port you choose does not matter.)



Configuring IP Addresses and Subnet Masks on the Hosts

Before we can communicate between the hosts we need to configure IP Addresses and Subnet Masks on the devices.

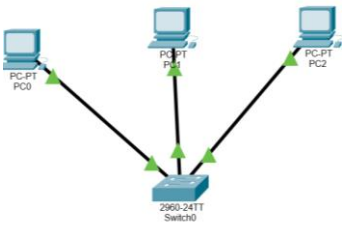
Host	IP Address	Subnet Mask
PC0	172.16.1.10	255.255.0.0
PC1	172.16.1.11	255.255.0.0
PC2	172.16.1.12	255.255.0.0
PC3	172.16.1.13	255.255.0.0

Wide Area Network (WAN)

WAN (Wide Area Network) is a type of computer network that covers a **large geographical area**, often connecting multiple LANs together across cities, countries, or even continents, using routers, leased lines, satellites, or the internet.
Examples include:

- A national banking network,
- Multinational company branches,
- Government communication systems, or
- The Internet (largest WAN).

Building the WAN – Connecting the Hosts to Switches
1st Building 2 Local Area Networks



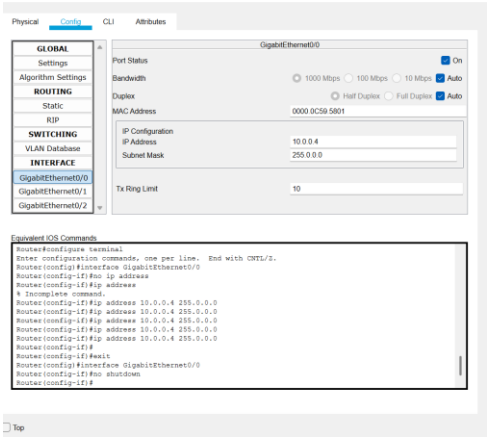
Adding a Router:

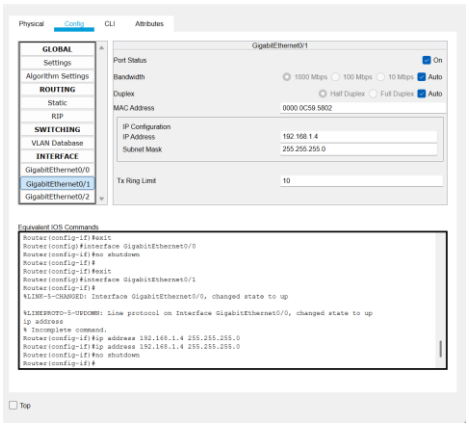
A **router** is a networking device that connects **two or more different networks** together (for example, a LAN to the Internet). It forwards data packets between networks using **IP addresses**.



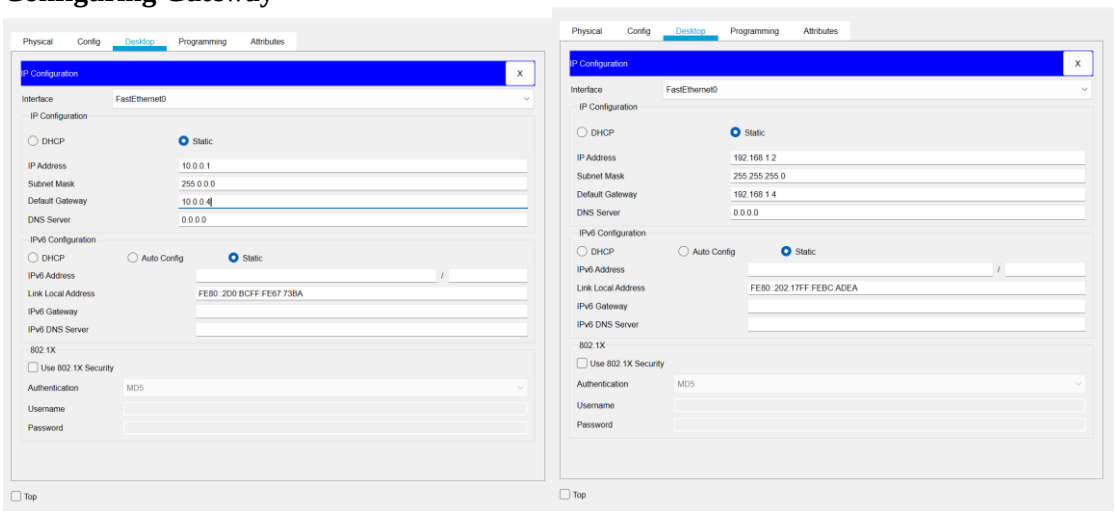
Configuring IP Addresses and Subnet Masks on the Hosts

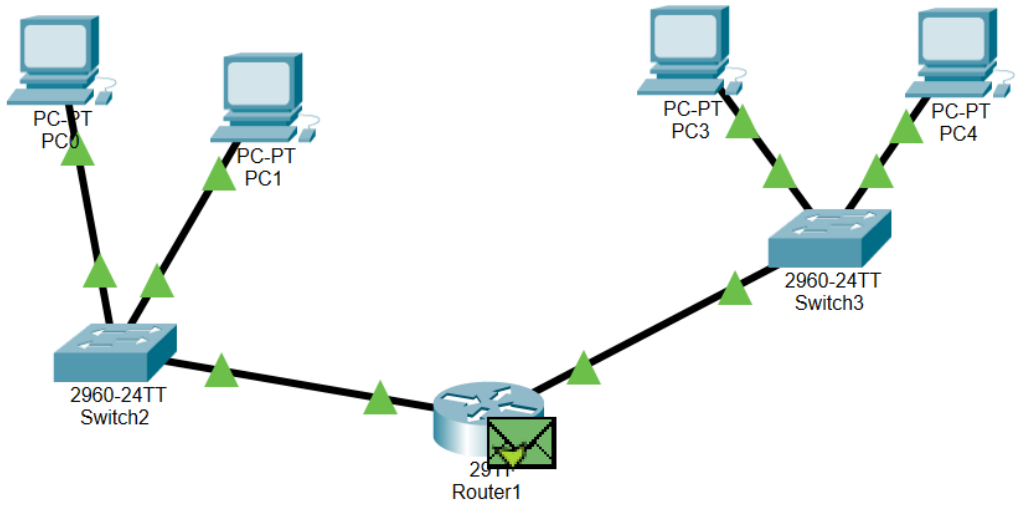
Host	IP Address	Subnet Mask
PC0	10.0.0.1	255.0.0.0
PC1	10.0.0.2	255.0.0.0
PC2	192.168.1.1	255.255.255.0
PC3	192.168.1.2	255.255.255.0





Configuring Gateway





Task:

Design a **WAN** in which **two separate LANs** are connected through **routers**. Each LAN should consist

of **4 PCs** connected via a switch. Configure IP addresses, connect the LANs using the routers, and test communication (ping) between PCs from different LANs.

Topology

A Network Topology is the arrangement with which computer systems or network devices are connected to each other. Topologies may define both physical and logical aspect of the network. Both logical and physical topologies could be same or different in a same network.
The types of computer network topologies are as follows:

1. **Bus Topology**
2. **Star Topology**
3. **Ring Topology**
4. **Mesh Topology**

Bus Topology

A bus topology is a network in which nodes are directly linked with a common half-duplex link. A host on a bus topology is called a station. In a bus network, every station will accept all network packets, and these packets generated by each station have equal information priority. A bus network includes a single network segment and collision domain.
Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

- To assign an IP address in PC0, click on PC0.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.
- Fill IPv4 address and subnet mask.

Configuring IP Addresses and Subnet Masks on the Hosts

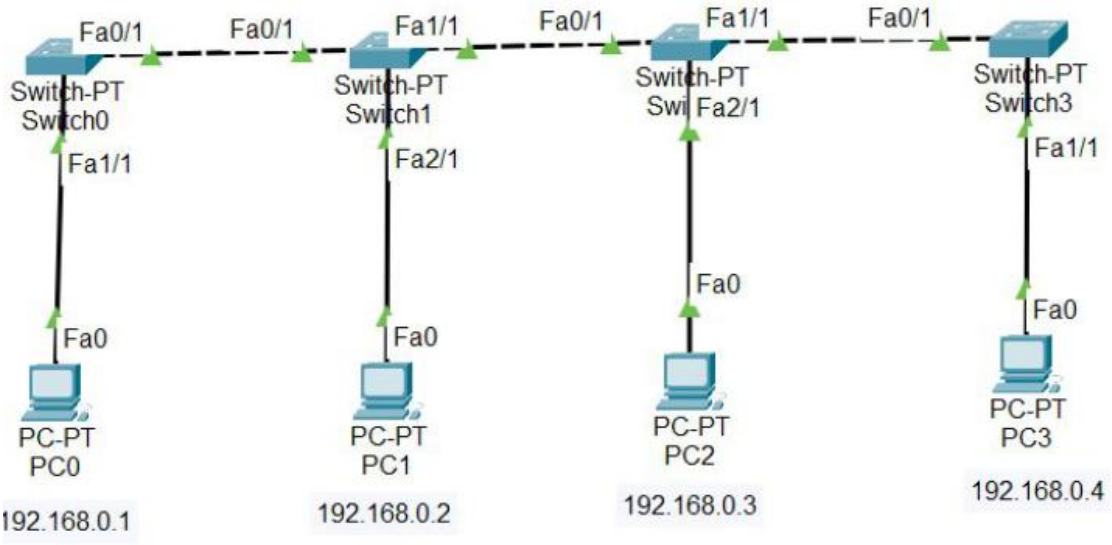
Before we can communicate between the hosts we need to configure IP Addresses and Subnet Masks on the devices.

IP Addressing Table:

S.NO	Device	IPv4 Address	Subnet Mask
	pc0	192.168.0.1	255.255.255.0
	pc1	192.168.0.2	255.255.255.0
	pc2	192.168.0.3	255.255.255.0
	pc3	192.168.0.4	255.255.255.0

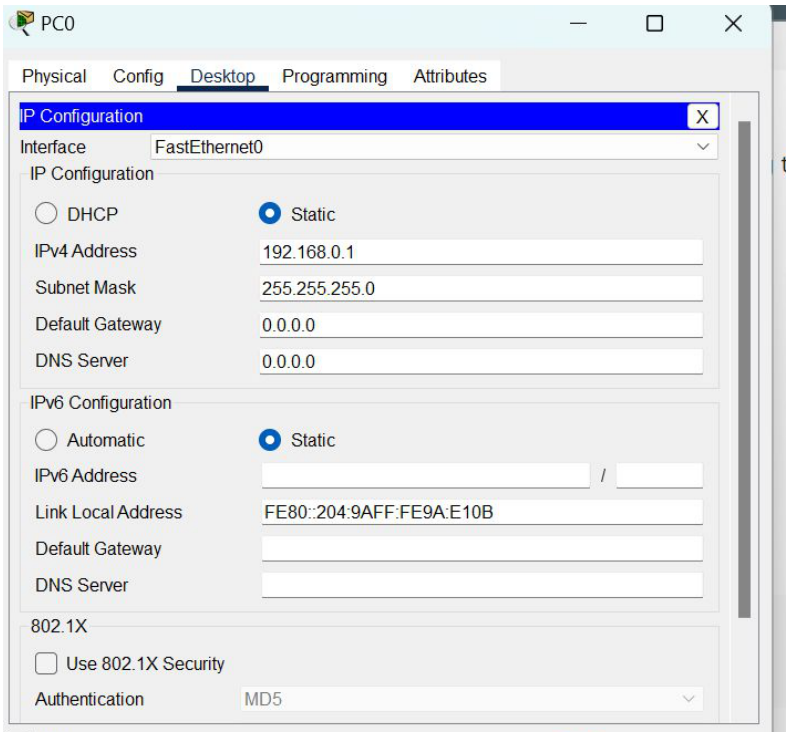
Then, create a network topology as shown below image:
Use an Automatic connecting cable to connect the devices with others.

Configure and Setup Bus Topology



Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

- To assign an IP address in PC0, click on PC0.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.
- Fill IPv4 address and subnet mask.

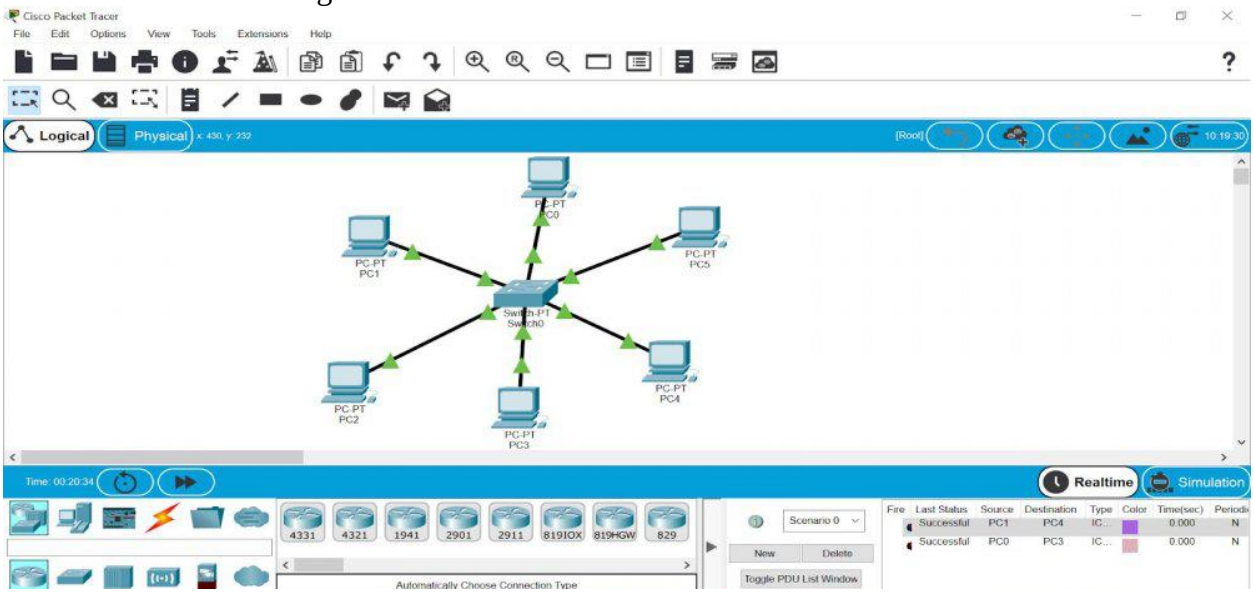


Simulation Result:

A simulation of the experiment is given below we have sent two PDU packets one targeted from PC0 to PC2 and another targeted from PC3 to PC1.

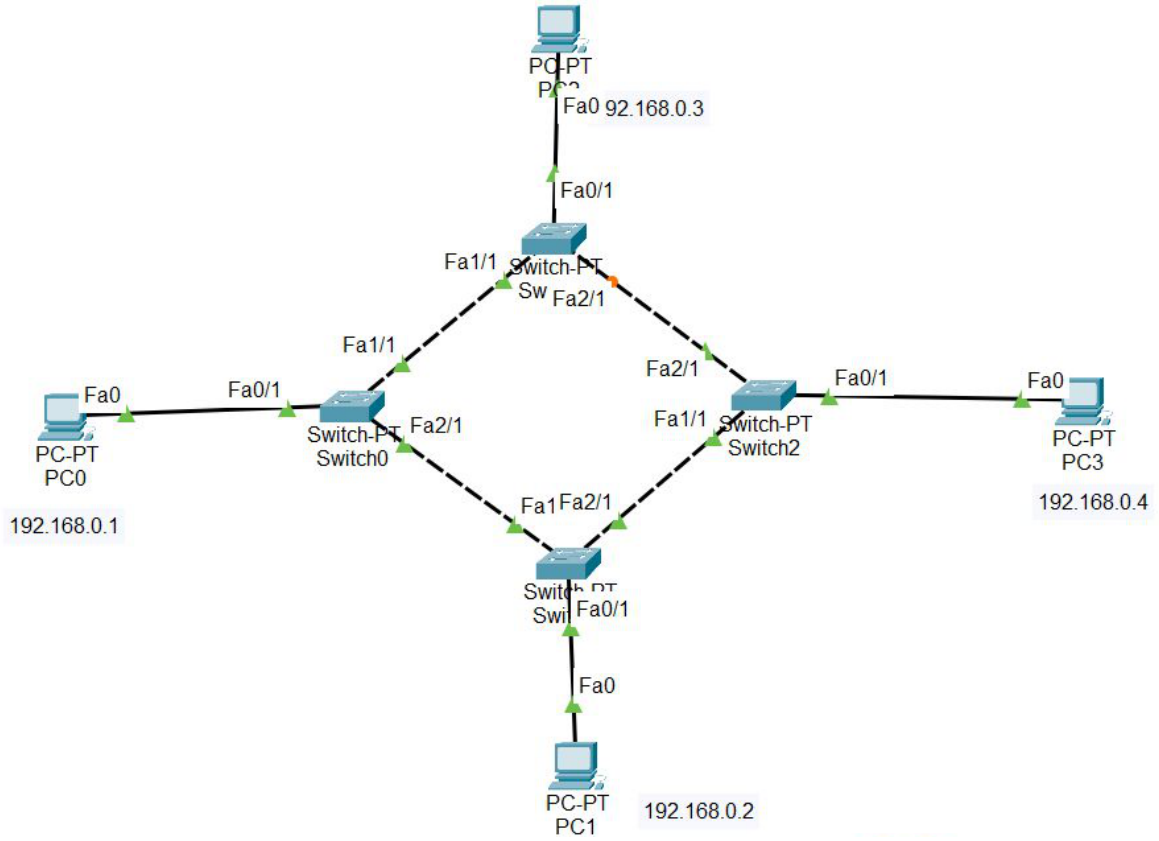
Start Topology

A star topology for a Local Area Network (LAN) is one in which each node is connected to a central connection point, such as a hub or switch. Whenever a node tries to connect with another node then the transmission of the message must be happening with the help of the central node. The best part of star topology is the addition and removal of the node in the network but too many nodes can cause suffering to the network.



Ring Topology

Ring topology is a kind of arrangement of the network in which every device is linked with two other devices. This makes a circular ring of interconnected devices which gives it its name. Data is usually transmitted in one direction along the ring, known as a unidirectional ring. The data is delivered from one device to the next until it reaches the decided destination. In a bidirectional ring, data can travel in either direction.



Mesh Topology

In the mesh topology of networking, each and every device sends its own signal to the other devices that are present in the arrangement of the network.

